William Caron-Bastarache

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Education

Université de Montréal — Montréal, QC

B.Sc. in Computer Science (Expected April 2028) — *2024 – Present*

- Relevant coursework: Software Engineering, Algorithms & Data Structures, Operating Systems, Probability & Statistics, Linear Optimization, Economics

Technical Skills

Programming: Python, C, Java, SQL, Bash, OCaml, JavaScript/TypeScript
Systems: Linux, POSIX, pthreads, CMake, gdb, Valgrind
Data & ML: PostgreSQL, Pandas, NumPy, scikit-learn, Matplotlib, psycopg
Frameworks & Web: FastAPI, REST APIs, asyncio, Docker, React, Next.js, HTML/CSS
Testing & Tools: pytest, JUnit, JaCoCo, PIT, GitHub Actions, Git, GitHub, Make, Maven, Claude Code
Languages: French (native), English (fluent)

Projects

Event-Driven Market Probability Engine (In Progress) | *PostgreSQL, Python, scikit-learn, Make, pytest* 2026

- Built a layered PostgreSQL warehouse and Python + SQL pipeline covering 15 ETFs and 32,000+ daily rows, ingested concurrently with asyncio and rebuilt from a single Make command.
- Implemented walk-forward validation across 5 expanding folds with a 60-day purge/embargo to prevent lookahead bias; trained logistic-regression models and stored predictions in PostgreSQL.
- Enforced point-in-time correctness with 36 pytest cases and 6 database invariants that re-derive each stored value from its definition, with each invariant validated by deliberately breaking it.
- Backtested three positioning rules net of transaction costs against a buy-and-hold benchmark (Sharpe 1.48); the strategies reached 1.22 and 0.66, confirming no exploitable edge after costs.

Building Operating System Components in C | *C, pthreads, POSIX, CMake, Valgrind, gdb* 2026

- Built a Unix-style shell supporting pipelines of arbitrary length and short-circuiting `&&`/`||` using fork, exec, dup2, and pipe.
- Developed a concurrent CPU scheduler with pthreads, mutexes, and condition variables; swept the time quantum across 8 values and benchmarked FCFS, SJF, and Round Robin on a 50-process workload, with no Valgrind-reported memory errors.
- Implemented virtual memory with 256 virtual pages, 32 physical frames, an 8-entry TLB, bitfield-packed page tables, FIFO eviction, dirty-page writeback, and instrumentation of TLB hit rate.
- Parsed FAT32 structures directly from raw bytes and used gdb to diagnose correctness and synchronization issues.

Implementing & Benchmarking Classic Algorithms | *Python, NumPy, Algorithm Analysis* 2026

- Benchmarked two Gale–Shapley implementations with the same $O(n^2)$ complexity, measuring a 3.8× runtime difference and identifying an $O(n^2)$ validation routine that contaminated the benchmark.
- Compared greedy and dynamic-programming approaches on an optimization problem; the dynamic program achieved a 3.2× faster runtime and a 4.7× higher objective value.
- Implemented and benchmarked an O(n log n) closest-pair algorithm, comparing empirical runtime scaling against theoretical complexity.

GraphHopper Software Quality Engineering | *Java, JUnit, JaCoCo, PIT, GitHub Actions* 2025

- Measured test coverage with JaCoCo and evaluated fault detection using PIT mutation testing, increasing mutation score and test strength by killing previously surviving mutants.
- Built a GitHub Actions quality gate that compares mutation-testing results against a persisted baseline and fails builds when mutation score regresses.

Experience

Incoming Data Scientist Intern – Optimization — Winter 2027

BRP — *Québec, Canada*

- Incoming Data Scientist intern applying optimization and quantitative methods to operational problems.

Lifeguard — 2022 – Present

City of Brossard — *Brossard, QC*